MILITARY SPECIFICATION

PELLET, RED PHOSPHORUS, FOR MORTAR CARTRIDGE 81MM SMOKE, RP, M819

This specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies for the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and quality assurance provisions for a smoke pellet used in 81MM Mortar Ammunition.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitations.

SPECIFICATIONS

MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, U S Army Armament Research, Development and Engineering Center, Attn AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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STANDARDS

FEDERAL

FED-S TD-313 - Material Safety Data Sheets, Preparation and the Submission of

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documenta, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified the issues shall be those in effect on the date of the solicitation.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT DRAWINGS (See 6.3)

9327838 Projectile, 81 MM Smoke, RP M819 Loading Assembly

9327876 Pellet, Smoke

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the document which is DoD adopted shall be that listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of a document not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Artillery

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2.3 Order of precedence. In the event of a conflict between the text of this specification and references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of-this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components The pellets shall comply with all requirements specified on drawing 9327876 and associated dwgs., and with all requirements specified in applicable specifications and standards The pellets shall have a uniform appearance of color and texture. Pellets shall not show evidence of crazing. pellets shall be free of cracks, pits, dirt, grease, and other foreign material that may adversely affect function. The handling methods used shall not be injurious to the pellets, nor shall any pellets be contaminated.

3.3 Pellet strength. Pellets shall withstand a compressive load of 6,000 psi when tested in accordance with 4.5.2 herein.

3.4 Burn test. Pellets shall weigh $14_g \pm 4_g$ after 3.5 minutes of burning when tested in accordance with 4. 5.3 herein.

3.5 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein. (See 4.3 and 6.2)

3.6 Workmanship.

3.6.1 Pellets. Pellets shall be free from foreign matter , gouges, dents, deformation or any similar defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or Purchase order, the contractor is responsible for the performance ofall inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements in sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated (See Table 1) .

Part description	Drawings	Quantity
Pellet, Smoke	9327876	25

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70550 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Pellet, Smoke				See Below 9327876
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for defects	25	100%	3.2	4.4.2.1
	Pellet strength	10	100%	3.3	4.5.2
	Burn test	10	100%	3.4	4.5.3

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of pellets shall contain:

a. Red phosphorus of one lot interfix number, from one manufacturer.

b. Sodium nitrate from one lot interfix number, from one manufacturer.

4.4.2 Examinination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Pellet, Smoke		9327876
			NEXT HIGHER ASSEMBLY 9327838

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Profile of pellet		0.40%	3.2	Gage
102	Height of pellet		0.40%	3.2	Gage
103	Pellet weight		0.40%	3.2	Gage
104	Compression strength	1/	100%	3.3	Gage
105	Burn test	1/	100%	3.4	Gage
106	Evidence of crazing or cracking		0.40%	3.2	Visual
Minor					
201	Presence of foreign matter		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

1/ Two samples shall be removed each 20 minutes of production.

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4.4.3 Testing Testing is described in the First Article and Quality Conformance Inspection tables.

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials. Compression testing equipment must be shielded and should allow for venting.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this specification is identified either directly or by reference, in the “Paragraph Reference/Inspection Method” column of the Quality Conformance Inspection tables herein. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all pellets have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Compression test. After completely curing, pellets shall be compressed between two flat parallel plates which completely cover the top and bottom face. The plates shall move together at 0.10 inches per minute. Failure to meet the requirement in 3.3 herein shall be cause for rejection of all material produced since the last successful test.

4.5.3 Burn test. Pellets will be placed on an approved scale in the center of a 2 feet diameter tube. Air at 56% ± 2% relative humidity, 72° ± 3°F shall be drawn into the tube at 600 cubic feet per minute. When fully cured, the pellet shall be oriented such that when either parallel face is on the scale surface a line which bisects the 900 angle of the pellet will be perpendicular to the direction of airflow over the pellet. The pellet shall be quickly ignited on the front-facing area and the ignition source quickly withdrawn from the air stream. Timing begins the moment the pellet is first ignited. Failure to comply with the requirement in 3.4 herein, shall be cause for rejection of all

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material produced since the last successful test. If the test is disturbed as indicated by a sharp change in the slope, the test may be repeated with another specimen representing material produced at the same time interval. Pellets must be fully cured prior to testing therefore identity of the pellets is required until the test is satisfied.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging of pellets for shipping shall be in accordance with 49 CFR 100-199. Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Alternate procedures to those contained in this specification shall be approved in advance by: Commander, USA ARDEC, ATTN: AMSMC-QAT-B (D), Dover, NJ 07801-5001.

6.2 Ordering data. See MIL-A-48078.

6.3 Drawings. Drawings listed in Section 2 of this specification under the headings U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal and U.S. Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.4 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit desings as required to Commander, U.S. Army Armament Research, Development and Engineering Center, (ARDEC) ATTN: AMSMC-QAT-I (D) , Dover, NJ 07801-5001.

6.5 Material safety data sheets. The contracting officer will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in appendix B of FED-STD-313.

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6.6 Subject term (key word) listing.

Ammunition, mortar
Inter fix
Pellet, smoke
Red phosporus
Sodium nitrate
Test, burn
Test, compression

Custodian:
ARMY-AR

Preparing activity:
ARMY-AR

(Project 1370-A197)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70550	PELLET, RED PHOSPHORUS, FOR MORTAR CARTRIDGE 81MM SMOKE, RP M81

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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